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The total displacement from 0 s to 40 s is given by

$$300\text{ m} + 300\text{ m} + 450\text{ m} - 200\text{ m} = +850\text{ m}.$$

Thus the particle's net displacement is along the positive x-direction.

The total distance travelled by the particle from 0 s to 40 s $= 300 + 300 + 450 + 200 = 1250$ m.

(c) The acceleration is given by the slope in the velocity-time graph. In the first 10 seconds the velocity has constant slope (constant acceleration). It implies that the acceleration a is from $v_1 = 0$ to $v_2 = 60\text{ m s}^{-1}$.

Hence $a = \frac{v_2 - v_1}{t_2 - t_1}$ gives

$$a = \frac{60 - 0}{10 - 0} = 6\,m\,s^{-2}$$

Next, the particle has constant negative slope from 15 s to 30 s. In this case $v_2 = 0$ and $v_1 = 60\,m\,s^{-1}$. Thus the acceleration at t = 20 s is given by $a = \frac{0 - 60}{30 - 15} = -4\,m\,s^{-2}$. Here the negative sign implies that the particle has negative acceleration.

EXAMPLE 2.32

If the position vector of the particle is given by $\vec{r} = 3t^2\hat{i} + 5t\hat{j} + 4\hat{k}$, Find the

a) The velocity of the particle at t = 3 s

b) Speed of the particle at t = 3 s

c) acceleration of the particle at time t = 3 s

Solution

(a) The velocity $\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$

We obtain, $\vec{v}(t) = 6t\hat{i} + 5\hat{j}$

The velocity has only two components $v_x = 6t$, depending on time t and $v_y = 5$ which is independent of time.

The velocity at t = 3 s is $\vec{v}(3) = 18\hat{i} + 5\hat{j}$

(b) The speed at t = 3 s is $v = \sqrt{18^2 + 5^2} = \sqrt{349} \approx 18.68\,m\,s^{-1}$

(c) The acceleration $\vec{a}$ is, $\vec{a} = \frac{d^2\vec{r}}{dt^2} = 6\hat{i}$

The acceleration has only the x-component. Note that acceleration here is independent of t, which means $\vec{a}$ is constant. Even at t = 3 s it has same value $\vec{a} = 6\hat{i}$. The velocity is non-uniform, but the acceleration is uniform (constant) in this case.

EXAMPLE 2.33

An object is thrown vertically downward. What is the acceleration experienced by the object?

Solution

We know that when the object falls towards the Earth, it experiences acceleration due to gravity g = 9.8 m s^{-2} downward. We can choose the coordinate system as shown in the figure.

